Depth Feature Extraction for Hyperspectral Image Small Sample Classification

Bing Liu, Xiaohui Chen, Zhixiang Xue, Pengqiang Zhang, Bing Zhang, and Jiaying Yue

***Abstract*— The problem of insufficient labeled samples has restricted the application of deep learning method in hyperspectral image (HSI) classification tasks. Fusion of remote sensing images from different sources such as HSI and LiDAR is a common strategy to improve the classification accuracy. However, obtaining multisource registered remote sensing images of the same area is time-consuming, which limits the application of multisource strategy in practice. Motivated by the recent success of large models in different fields, we propose to extract depth information from large models and fuse it with HSIs to improve the small sample classification accuracy. Specifically, we use the pretrained foundation large model to estimate the depth information of HSIs as the depth features, and then input the original spectral features and depth features into the support vector machine (SVM) to complete the classification. In order to further improve the classification accuracy, we propose to use the sliding window method to extract the depth features of different bands, so as to obtain more rich depth features. A large number of classification experiments on six benchmark datasets verify the effectiveness of the proposed method.**

***Index Terms*— Deep learning, feature extraction, few-shot classification, foundation large model.**

I. Introduction

HYPERSPECTRAL sensors detect objects through a large number of different wavelength bands of the electromagnetic spectrum to obtain hyperspectral images (HSIs) [1]. The information collection range of the hyperspectral sensor can cover the information of the entire electromagnetic spectrum, far exceeding the visible light band that can be distinguished by the human eye [2]. Hence, the HSI contains more information that can be interpreted, which provides rich information for fine ground object identification [3]. However, the problem of small samples brings great challenges to the HSI classification.

Feature extraction could produce more discriminative features, which is the most commonly used method to deal with the problem of small sample in HSI classification tasks [4]. Initially, the feature extraction of HSIs only considers the spectral dimension, which ignores the data characteristics of the spatial and spectral integration of HSIs. In order to make full use of the spatial–spectral information provided by HSIs, the researchers proposed a series of spatial–spectral feature extraction methods [5]. Compared with the use of spectral dimension features alone, these spatial feature extraction methods combined with shallow classifiers such as support vector machine (SVM) and random forest greatly improve the HSI small sample classification accuracy [6]. For example, He et al. [7] design a spectral–spatial-range Gabor filtering to restrain edge interference from disturbing local spectral–spatial harmonic components, which achieves excellent performances. Li et al. [8] design a multidimensional local binary pattern based on Clifford algebra for HSI classification, which could extract spatial–spectral features from multiple dimensions. Falco et al. [9] combine morphological attribute profiles and independent component analysis for spatial–spectral classification of HSIs. The disadvantage of the abovementioned spatial–spectral feature extraction method is that experts need to manually design feature extraction rules, which leads to the final classification accuracy relying heavily on expert experience.

The deep learning method could automatically learn and extract abstract features suitable for downstream tasks from HSIs [10]. The rise of deep learning method effectively solves the problem of manually designing feature extraction rules in HSI classification tasks. Therefore, in the past five years, the classification model represented by the convolutional neural network (CNN) has received extensive attention and exploration in the task of HSI classification [11]. For example, Yue et al. [12] first apply CNN to HSI classification. Yu et al. [13] design a simplified 2-D–3-D CNN architecture for HSI classification in the foundation of spatial–spectral fusion. In order to make full use of the spatial–spectral information contained in HSIs, researchers propose to treat HSIs as time-series data along the spectral dimension, and then use the recurrent neural network (RNN) to model the relationship between image features and labels [14]. The attention mechanism could enhance the fitting ability of deep learning model, so it is widely used in HSI classification tasks. For example, Song et al. [15] design an interactive spectral–spatial transformer for HSI classification. Sun et al. [16] design a memory-augmented spectral–spatial transformer for HSI classification.

Although the deep learning method does not need to design feature extraction rules manually, it needs a large number of labeled samples to optimize thousands of parameters [17].

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It is time-consuming and laborious to obtain labeled samples in HSIs, which undoubtedly limits the application of deep learning method in HSI classification tasks [6]. Semi-supervised learning is a method that can make full use of unlabeled samples to improve classification accuracy, so it is also one of the most effective methods to alleviate the problem of small samples [18]. In order to improve the classification accuracy of small samples, researchers have explored a large number of semi-supervised learning methods suitable for the characteristics of HSIs. For example, Song and Yang [19] design a semi-supervised robust self-ensemble network for HSI classification, which achieves high classification accuracy on the international common test datasets. Chen et al. [20] design a semi-supervised learning method for HSIs based on pseudo-label strategy, which achieves high classification accuracy under the condition of small samples. At the same time, researchers propose to use meta-learning strategy to further improve the classification accuracy of small samples. Meta learning can effectively improve the classification accuracy by learning small sample classification scenarios. For example, Zhang et al. [21] design a Bayesian meta-learning method for few-shot HSI classification. Semi-supervised learning and meta-learning methods still need labeled samples to train the deep learning model. By constructing self-consistent learning tasks, self-supervised learning can learn features suitable for small sample classification tasks from unlabeled samples. Therefore, self-supervised learning is widely used in HSI classification tasks with limited samples [22]. Liu et al. [23] construct a self-supervised learning method suitable for small sample classification through the constraint of spectral dimension, which greatly improved the small sample classification accuracy on four benchmark HSI datasets.

The above methods deal with the small sample problem in the HSI classification task from the perspective of model and learning strategy. Multisource remote sensing image collaborative classification strategy improves the classification accuracy of HSIs from the perspective of data use. It improves the classification accuracy by introducing new information from different sources of remote sensing images [24]. A large number of studies and practices have proved the effectiveness of the appropriate use of multisource remote sensing images. Therefore, multisource remote sensing image collaborative classification strategy has been widely studied in HSI classification tasks [25].

As for the collaborative classification strategy of multisource remote sensing images, the common types are HSI and LiDAR data [26], HSI and multispectral data [27], and HSI and synthetic aperture radar (SAR) data [28]. Early methods use feature extraction technology to extract features from different types of remote sensing data, and then fuse the extracted features as the input of classifier [29]. In recent years, researchers have explored to use deep learning models for collaborative classification of multisource remote sensing images. The most convenient way is to use the multibranch network architecture to construct an end-to-end classification method. Specifically, the multibranch network extracts different types of features using multiple network branches, and then fuses the learned features to complete the classification task. As for the multibranch network, the original CNN [30], residual network [31], and capsule network [32] could be used to extract different types of features from HSIs and digital surface model (DSM) data, respectively. Similarly, we can use a three-branch network architecture to extract the spatial features, the spectral features, and elevation features [33].

Although multisource remote sensing image collaborative classification strategy can improve the classification accuracy, the acquisition of multisource registered remote sensing image of the same region is time-consuming and laborious, which limits the practical application of this strategy. In addition to obtain real multisource remote sensing images, monocular image depth estimation can also provide additional information for classification tasks. In recent years, monocular image depth estimation methods based on deep learning have been widely studied. According to whether the real depth value is used as the label, the deep learning-based methods are divided into supervised learning, unsupervised learning, and semi-supervised learning [34]. The supervised method uses CNN to directly establish the mapping between color pixels and depth values, and uses the real depth values as labels [35]. The real depth value required by the supervised method in network training is difficult to obtain in practical application. Therefore, in order to deal with this problem, many researchers try to use unsupervised deep learning method for monocular depth estimation task. The unsupervised method always uses the self-supervised strategy [36]. It transforms the depth estimation problem into other visual problems (such as image reconstruction problem). The semi-supervised learning method improves the generalization ability of the model by using a small amount of labeled data and a large amount of unlabeled data [37].

In the past two years, the pretrained large language model represented by ChatGpt has achieved great success in natural language processing tasks [38]. Through pretraining with massive texts and fine-tuning with a small amount of supervised data, it has shown strong zero-shot generalization ability in different natural language processing tasks. Inspired by this, researchers proposed to build a foundation large model for image classification, semantic segmentation, object detection, and other visual tasks. Segment anything model (SAM) [39] is one of the most typical large vision models. The goal of SAM is to segment any object in the image by using the easily available prompt information. Through sufficient training on 11 million images and more than 1 billion masks, SAM has mastered the general concept and knowledge of objects, and even has good zero-shot generalization ability for unseen objects and unseen scenes.

The large vision model shows a strong generalization ability in general visual tasks. Inspired by this, we propose to use a large vision model to extract depth information as additional features for collaborative classification with HSIs. The main contributions of this work are summarized as follows.

1) We first propose to use the large vision model to obtain the depth feature information of HSIs for collaborative classification. The proposed method adopts the multisource fusion strategy to improve the classification accuracy of HSIs. More importantly, it does not need

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additional remote sensing images, which is plug-and-play in nature.

2) Motivated by the multifeature maps of CNN, we propose to use the sliding window method to obtain more abundant depth feature information along the spectral dimension, which greatly improves the accuracy of collaborative classification, especially for small sample conditions.
3) To verify the effectiveness of the proposed method, a large number of classification experiments are carried out on four HSI benchmark datasets and two multisource datasets. The experimental results show that the proposed method can obtain more competitive results than the comparison method.

The remainder of this article is organized as follows. In Section II, we elaborate on the proposed method. In Section III, we conduct classification experiments and corresponding analysis. In Section IV, we summarize this article and give the conclusion.

II. Proposed Method

The key of our method is to build a large vision model to estimate the depth information of any image. Inspired by the recent monocular image depth estimation method, we follow the idea of [40] to build a foundation vision model. Then, we train the model by semi-supervised learning method, so as to ensure that more unlabeled data can be used, thus ensuring the generalization of the model. Finally, we use the pretrained model to extract the depth information of different bands along the spectral dimension, and combined it with support vector machine to complete small sample classification. The proposed method will be described in detail in the following.

A. Model Architecture

As shown in Fig. 1, the monocular image depth estimation model is composed of an encoder and a decoder. We use the classic vision transformer (ViT) [41] as the encoder. ViT evenly divides the input image into 16×16 patches. Then, a linear projection layer is applied to the patches to obtain visual tokens. In order to take into account the spatial information, ViT encodes the spatial position of each visual token as a feature vector and fuses it with the visual token. Finally, the fused visual tokens pass through several blocks. ViT can be formally described as follows:

$$\begin{aligned} z_0 &= \left[\boldsymbol{x}_p^1, \boldsymbol{x}_p^2, \ldots, \boldsymbol{x}_p^N\right] + \boldsymbol{E}_{\text{pos}} \\ z'_l &= \text{MSA}(LN(z_{l-1})) + z_{l-1}, \quad l = 1, \ldots, L \\ z_l &= \text{MLP}(LN(z'_l)) + z', \quad l = 1, \ldots, L \end{aligned} \tag{1}$$

where $\boldsymbol{x}_p = \left[\boldsymbol{x}_p^1, \boldsymbol{x}_p^2, \ldots, \boldsymbol{x}_p^N\right]$ is the patch embeddings, $\boldsymbol{E}_{\text{pos}}$ is the position encoding feature matrix, the dimension of $\boldsymbol{E}_{\text{pos}}$ is the same as $\boldsymbol{x}_p$, MSA is a multihead self-attention layer, LN is a normalization layer, and MLP is a multilayer perceptron layer. The size of the input image is $H \times W \times C$, the size of $\boldsymbol{x}_p$ is $N \times (P^2 \times C)$, C is the number of image channels, P is the size of patch, and $N = H \times W/P$. Each block of ViT is composed of a multihead attention layer, an MLP layer, two normalization layers, and two residual connections. Different versions of ViT are formed according to the number of block layers, hidden layer size, MLP size, and other parameters. The commonly used ViT-small, ViT-Base, and ViT-Large parameter configurations are shown in Table I.

TABLE I
Details of Three Variants of ViT

Model	Block	Hidden Size	Heads	Params
ViT-Small	12	384	6	22.1M
ViT-Base	24	768	12	86.6M
ViT-Large	32	1024	16	304.4M

TABLE II
Parameter Configuration of the Decoder

Layers	kernel size	stride	padding	out channels
Conv	1	1	0	256
ConvTranspose	4	4	0	128
ConvTranspose	2	2	0	32
Conv	3	2	1	1
Interpolate	-	-	-	-

The decoder consists of two convolution layers, two deconvolution layers, and an upsampling layer. Its final output is a depth map of the same size as the original image. The specific parameter configuration of the decoder is shown in Table II. The convolution layer (Conv) is used to transform the feature map, the deconvolution layer (ConvTranspose) is used to upsample the feature map, and finally, the interpolation layer samples the output depth map to the original image size.

B. Monocular Depth Estimation

The amount of labeled data for monocular image depth estimation is limited. Compared with the limited labeled data, it is easier to obtain a large amount of open-source unlabeled data. In order to ensure the generalization ability of the model, semi-supervised learning method is used to train the model. In addition, to improve the accuracy of depth estimation, the feature consistency constraint is introduced. The specific loss function is as follows:

$$L = L_l + L_u + L_{\text{feat}} \tag{2}$$

where L_l, L_u, and L_{feat} are the supervised loss, unsupervised loss, and feature consistency constraint, respectively. L_l is formalized as follows:

$$\begin{aligned} L_l &= \frac{1}{HW}\sum_{i=1}^{HW}\left|\hat{d}_i^* - \hat{d}_i\right| \\ \hat{d}_i &= \frac{d_i - t(d)}{s(d)} \\ t(d) &= \text{median}(d) \\ s(d) &= \frac{1}{HW}\sum_{i=1}^{HW}|d_i - t(d)| \end{aligned} \tag{3}$$

where $\hat{d}_i^*$ and $\hat{d}_i$ are the scaled and shifted versions of the prediction and ground truth, respectively. $t(d)$ and $s(d)$ are

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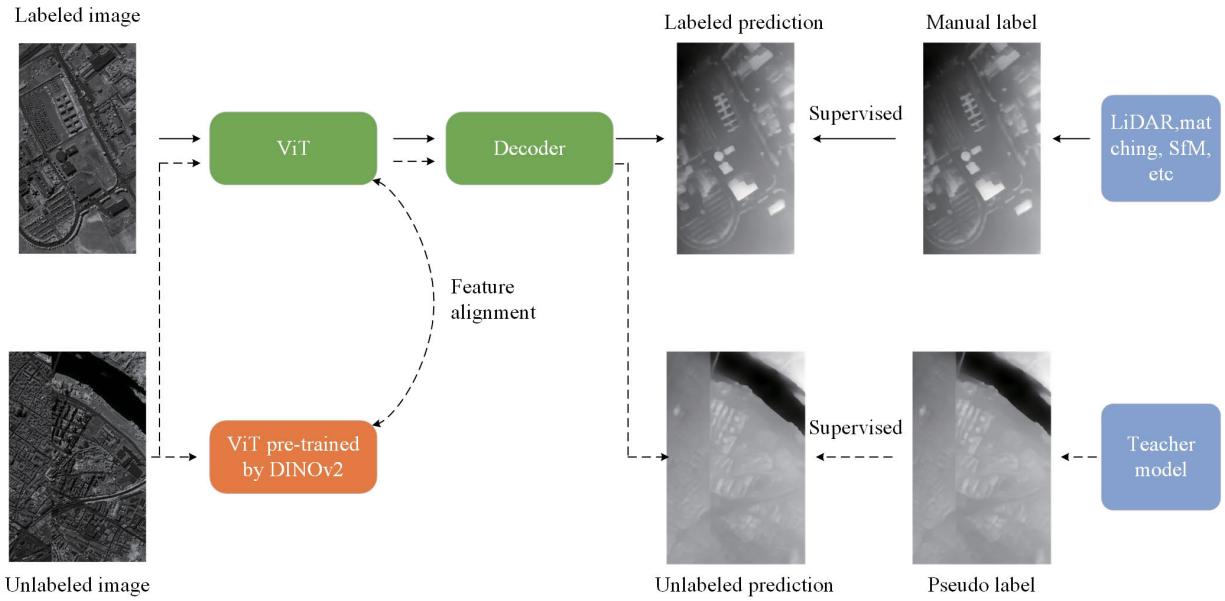


Fig. 1. Pipeline of the proposed method.

used to align the prediction and ground truth to have zero translation and unit scale.

First, the labeled data are used to train a teacher network T based on the supervised loss function. Then, T is applied to the unlabeled image to obtain the corresponding pseudo-labels. The student network S is trained by the labeled data and unlabeled data. The direct use of pseudo-labels to calculate the unsupervised loss has no obvious effect on improving the performance of the model. In order to deal with this problem, CutMix [42] is applied to the unlabeled image when calculating the unsupervised loss. By introducing CutMix to set more challenging optimization goals, model performance could be improved. CutMix needs to randomly select a pair of images I_a and I_b, and the enhanced image I_{ab} is

$$I_{ab} = I_a \odot M + I_b \odot (1 - M) \tag{4}$$

where M is a binary mask containing a rectangular region, and the rectangular region is identified as 1. We follow the original CutMix settings, and the two images are randomly selected. It is not required that the two images are in the same category.

The calculation of L_u is divided into two parts according to the mask M. The specific calculation is as follows:

$$\begin{aligned} L_u &= L_u^M + L_u^{1-M} \\ L_u^M &= \frac{\sum M}{HW} \sum_{i=1}^{H_M W_M} \rho(S(I_{ab}) \odot M, T(I_a) \odot M) \\ L_u^{1-M} &= \frac{\sum M}{HW} \sum_{i=1}^{H_M W_M} \rho(S(I_{ab}) \odot (1 - M), T(I_a) \odot (1 - M)) \end{aligned} \tag{5}$$

where H_M and W_M are the height and width of the rectangular area in the mask M, respectively, T is the teacher network trained by the labeled data, S is the student network to be trained, and $\rho(d_i^* - d_i) = |\hat{d}_i^* - \hat{d}_i|$ is similar to the calculation of supervision loss.

DINOv2 [43] performs well in image classification, semantic segmentation, object detection, and other tasks. Therefore, in the training process, the loss of feature consistency is used to guide the depth information output by the model to keep consistent in the same object area. The loss of feature consistency is calculated as follows:

$$L_{\text{feat}} = 1 - \frac{1}{H_f W_f} \sum_{i=1}^{H_f W_f} \cos(f_i, f_i') \tag{6}$$

where H_f and W_f are the height and width of the feature map, respectively, f_i is the feature map output by the encoder to be trained, f_i' is the feature map output by the DINOv2 encoder, and $\cos(\cdot, \cdot)$ is the cosine similarity between two feature vectors.

Compared with the traditional deep learning model, the generalization ability of large language model and large vision model mainly comes from more parameters and a wider range of data. Therefore, in order to ensure the generalization of the monocular image depth estimation model, the training data including 1.5M labeled images and 62M unlabeled images are constructed based on the open-source data in [40].

C. Depth Extraction Along the Spectral Dimension

After massive data training, the model can well estimate the depth information of any image. A simple idea is to input the visible light band of an HSI into the model to obtain depth information, and then carry out the collaborative classification of HSI and depth information. But considering that our goal is to accurately distinguish different objects, the absolute accuracy of depth estimation has little effect on the final classification accuracy. Therefore, we need to ensure the consistency of depth information of the same object and

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the difference of depth information between different objects. The consistent in the same object areas and the difference in different object areas are mainly realized by optimizing the loss function shown in (2). The loss function includes supervised loss function L_l, unsupervised loss function L_u, and feature consistency constraint L_{feat}. Among them, L_l supervises the training of the whole model by aligning the predicted relative depth value on the labeled data with the real absolute depth value. In order to improve the generalization of the model, the unsupervised loss function L_u is introduced. The main effect of feature consistency constraint loss term L_{feat} is to further ensure the consistency of depth information for the same object and the difference of depth information between different objects. When calculating the feature consistency loss constraint, we frozen the parameters of DINOv2 to align the features of the model to be trained with the features of DINOv2. DINOv2 can be generalized to semantic segmentation, object detection, and other visual tasks without any fine-tuning. Among the features extracted by DINOv2, the features corresponding to the same semantic region will approach the same, and the features corresponding to different semantic regions will be different. Therefore, using feature consistency loss can guide the depth information output by the model to remain consistent in the same object areas and the difference in different object areas. The effect of the feature consistency loss term is verified in the following experiment. After the feature consistency loss term is applied, the depth information consistency in the same object region is better, and the depth information difference between different object regions is greater.

Only using visible light band will ignore the spectral advantages of HSIs. Inspired by the idea of multifeature maps in CNN, we treat the depth information output by the model as depth features. As shown in Fig. 2, the sliding window method is used to extract the depth features of different bands along the spectral dimension. Specifically, HSIs are divided into several three-band images along the spectral dimension as the input of the pretrained foundational model, and each three-band image will pass through the model to obtain its corresponding depth information. We stack these depth images as depth features. After the training of the model in Fig. 1 is completed, it can be regarded as a feature extractor. HSI can be decomposed into a series of three-band images along the spectral dimension. When each three-band image is input into the model shown in Fig. 1, a single-band depth information map will be obtained. The depth features shown in Fig. 2 are obtained by stacking the single-band depth information maps corresponding to different three-band images. In the process of band division, we use the sliding window method. Take the University of Pavia (UP) data as an example. This data has 103 bands. The stride of the sliding window method is set to be 1. We will get 101 three-band images. Input these images into the model, respectively. We will get 101 depth maps, and stack these depth maps to form 101-D features.

HSIs contain rich spectral and spatial information, so we fuse the extracted depth features with the original spectral features. We adopt a simple but effective feature fusion method, that is, directly concatenating the separately normalized features. This feature fusion method is widely used in HSI classification tasks and has been proven to be effective. Finally, we input the fused features into a support vector machine classifier to complete the classification.

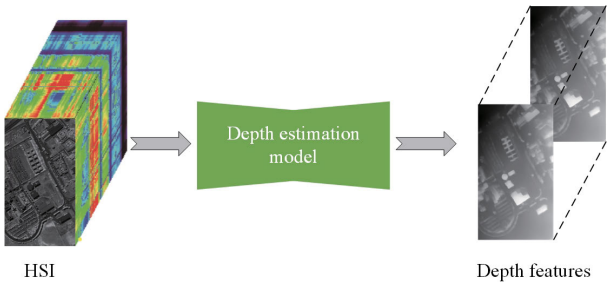


Fig. 2. Pipeline of depth extraction along the spectral dimension.

TABLE III
DETAILS OF THE HSIS. UP, PC, SA, IP, GROUND SAMPLE DISTANCE (GSD) (M), SPATIAL SIZE (PIXEL), SPECTRAL RANGE (NM), AIRBORNE VISIBLE INFRARED IMAGING SPECTROMETER (AVIRIS), AND REFLECTIVE OPTICS SYSTEM IMAGING SPECTROMETER (ROSIS)

	HSIs			
	UP	PC	SA	IP
Spatial size	610 × 340	1096 × 715	512 × 217	145 × 145
Spectral range	430-860	430-860	400-2500	400-2500
No. of bands	103	102	204	200
GSD	1.3	1.3	3.7	20
Sensor type	ROSIS	ROSIS	AVIRIS	AVIRIS
No. of classes	9	9	16	16
Labeled samples	42776	148152	54129	10249

III. EXPERIMENTAL RESULTS

A. Datasets

Four international benchmark datasets, UP, Pavia Center (PC), Salinas (SA), and Indian Pines (IP), are selected for classification experiments to verify the effectiveness of our method. The details of the four datasets are shown in Table III. HSI classification tasks usually need to deal with small sample scenes. In order to verify the effectiveness of the proposed method for small samples, five labeled samples per class are randomly selected as training samples, and the remaining samples as test samples.

To further verify the effectiveness of the proposed method, we selected Houston 2013 and Trento datasets for multisource remote sensing image classification experiments. The Houston 2013 and Trento, which are composed of registered HSI and LiDAR, are commonly used multisource remote sensing image classification benchmark datasets. We compare the classification results of the proposed method with the real multisource remote sensing image classification results, which can further verify the effectiveness of the proposed method. Details of Houston 2013 and Trento are shown in Table IV.

B. Experimental Settings and Ablation Study

After the depth feature extraction, we use radial basis function-support vector machine (RBF-SVM) to complete

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TABLE IV
DETAILS OF THE HSIS AND LIDAR

	Hoston 2013		Trento	
	HSI	LiDar	HSI	LiDar
Spatial size	349 × 1905	349 × 1905	166 × 600	166 × 600
Spectral range	380-1050	-	402.89-989.09	-
No. of bands	144	1	63	1
GSD	2.5	2.5	1.0	1.0
Sensor type	ITRES-CASI1500	DSM	AISA Eagle	Optech ALTM 3100EA
No. of classes	15	15	6	6
Labeled samples	15029	15029	30214	30214

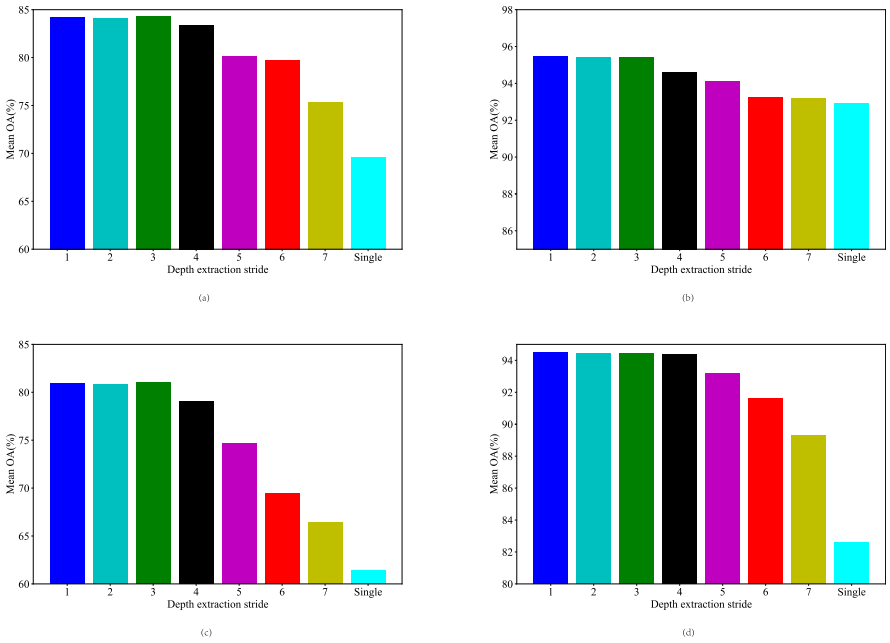


Fig. 3. Classification results with different depth extraction strides. (a) UP. (b) PC. (c) IP. (d) SA.

the classification. In all experiments, a grid search strategy is used to determine the parameters from a wide range $(2^{-2}, 2^{-1}, \ldots, 2^{7})$.

The depth extraction stride along the band dimension of the proposed method will greatly affect the classification results. To analyze this parameter, we performed experiments on four HSIs. We set the strides as 1, 2, 3, 4, 5, 6, and 7, respectively. In order to facilitate comparative analysis, we also use three visible light bands to extract depth information and carry out classification experiments, which are marked as "single." The classification results of the four hyperspectral datasets are shown in Fig. 3. The classification accuracy in Fig. 3 is the average of ten experimental results. According to the experimental results in Fig. 3, it can be seen that the classification accuracy will be greatly reduced when the depth information is extracted in the three visible bands. The extraction of depth information according to a certain stride will improve the classification accuracy, and the smaller the stride, the higher the classification accuracy. This is because the proposed method considers the depth information extracted by the model as features, and using more bands to extract the depth information will provide more discriminant features for the classifier. The proposed method does not depend on the absolute accuracy of depth information extraction, because as long as the depth information of different objects can show differences, it will help to improve the final classification accuracy. We find that the stride setting of 1, 2, and 3 has little effect on the final classification accuracy. The smaller the stride, the longer the feature extraction time and classification time will increase. We finally set the stride of depth feature extraction to 3.

Using ViT with different parameters to extract depth information will also affect the final classification accuracy. We tested ViT-small, ViT-base, and ViT-large on four HSIs, and the box plots of the corresponding classification results are shown in Fig. 4. According to the experimental results in Fig. 4, it could be found that the model with more parameters is helpful to improve the final classification accuracy because the model with larger parameters has stronger generalization ability. Finally, we choose ViT-large as the backbone network of the depth information extraction model.

Using multiple sample data augmentation during training is helpful to improve the model's generalization ability. We tested three different data augmentation methods, SamplePairing, MixUp, and CutMix. The experimental results of the three data augmentation methods are shown in Fig. 5. For the

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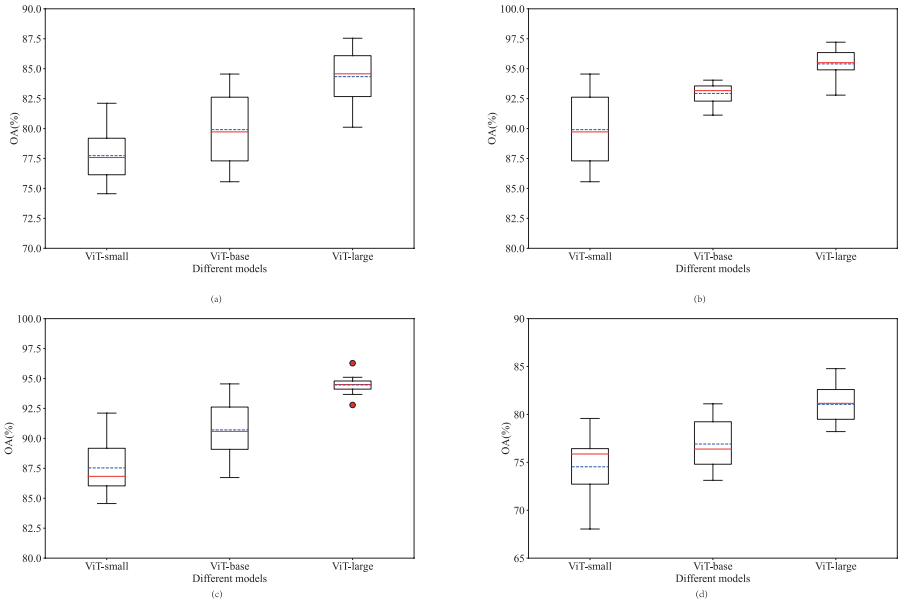


Fig. 4. Classification results with different backbones. (a) UP. (b) PC. (c) SA. (d) IP.

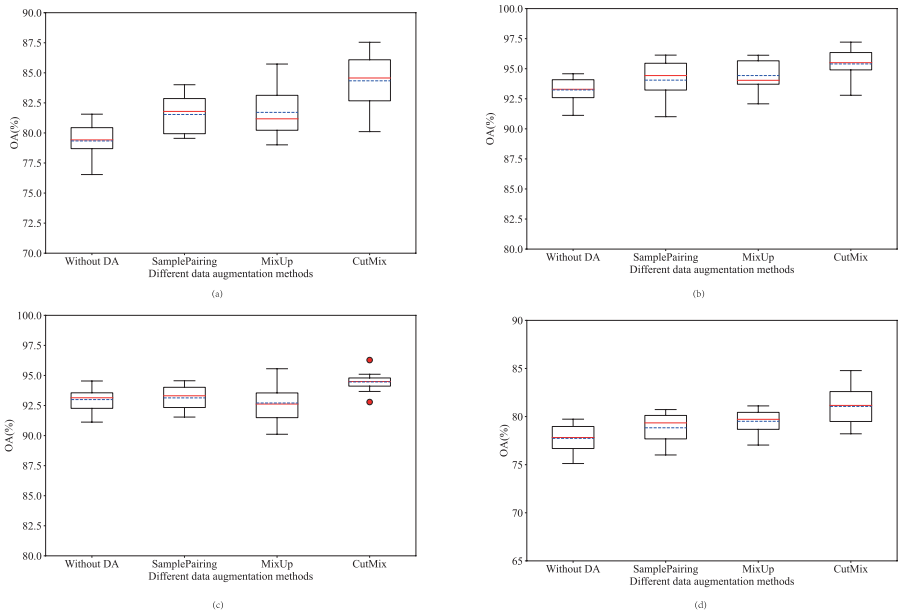


Fig. 5. Classification results with different data augmentation methods. (a) UP. (b) PC. (c) SA. (d) IP.

convenience of comparative analysis, we provide classification results without any data augmentation. From the experimental results in Fig. 5, it can be found that using multiple sample data augmentation helps to improve the classification accuracy of the HSIs. In addition, CutMix shows the most significant improvement. Considering that using multiple data augmentation strategies will increase training costs, we only used CutMix during the model training process.

To analyze the impact of using different similarity measurement functions in (6) on the final classification accuracy, we tested the Euclidean distance, correlation coefficient, and cosine similarity separately. The experimental results of different metric functions are shown in Fig. 6. According to the experimental results in Fig. 6, we found that different similarity measurement functions have little impact on the final classification accuracy. We ultimately chose the cosine similarity with the highest average of ten classification accuracies to measure the consistency of different features.

In order to verify the effectiveness of feature consistency loss in (6), we trained the depth prediction model using feature consistency loss and the depth prediction model without feature consistency loss for comparative analysis. Taking a three-band image of PU as input, the depth map predicted by the two models is shown in Fig. 7. According to the visualization results in the figure, the depth information of the same object area has better consistency in the predicted depth map outputted by the model with feature-consistent loss. The separability of different object categories is better

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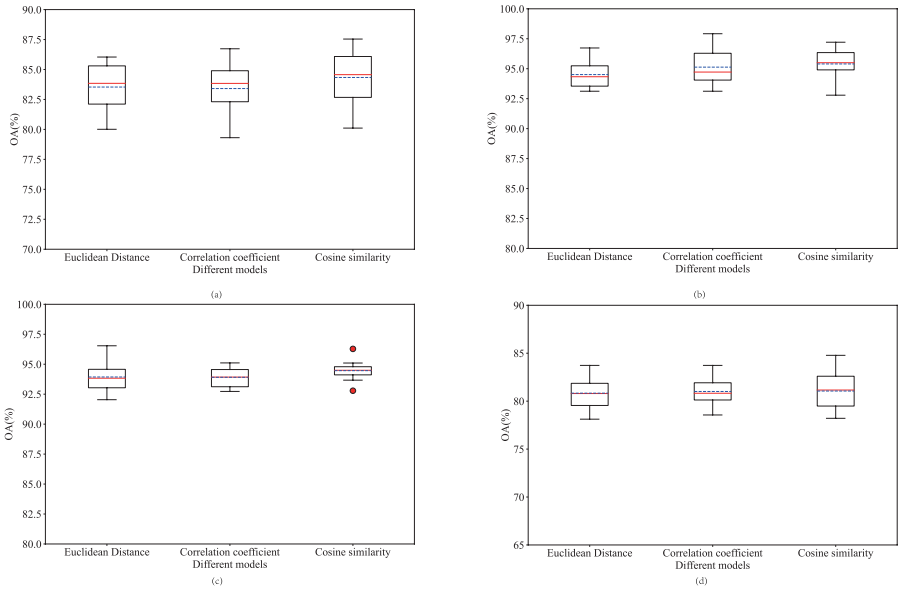


Fig. 6. Classification results with different similarity measurement functions. (a) UP. (b) PC. (c) SA. (d) IP.

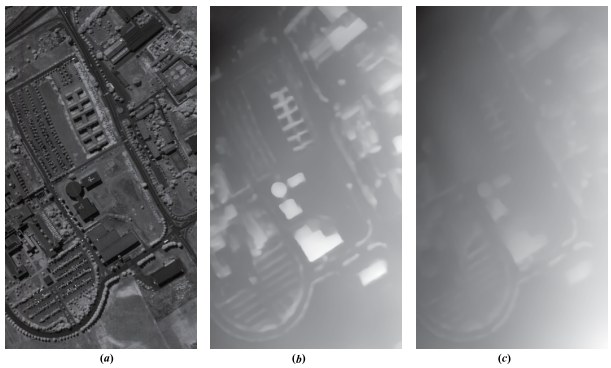


Fig. 7. Classification results with different similarity measurement functions. (a) Input image. (b) Depth prediction with feature consistency loss. (c) Depth prediction without feature consistency loss.

in the predicted depth map outputted by the model with feature-consistent loss. This is because, in the process of applying the feature-consistent loss training model, we have frozen the parameters of DINOv2 to align the features of the model to be trained with the features of DINOv2. DINOv2 can be generalized to semantic segmentation, object detection, and other visual tasks without any fine-tuning. Therefore, among the features extracted by DINOv2, the features corresponding to the same semantic region will approach the same. Therefore, using feature consistency loss can guide the depth information output by the model to remain consistent in the same object areas.

C. Comparison With Small Sample Method

In order to verify the effectiveness of the proposed method for small samples, eight different classification methods including 3D-CNN [44], EMP + GCN [45], DFSL + SVM [6], RN-FSC [46], DCFSL [47], UM^{2}L [22], ADAFSL [48], and CTF-SSCL [49] are selected for comparison. The 3D-CNN can directly take the data cube as the input and can better mine the spatial–spectral features to improve the classification accuracy. Therefore, 3D-CNN is selected as a test benchmark. EMP + GCN can use a large number of unsupervised samples to improve the accuracy of small sample classification in HSIs. As for 3D-CNN and EMP + GCN, five labeled samples per class are randomly selected as the supervised training samples. DFSL + SVM, RN-FSC, DCFSL, and UM^{2}L are meta-learning-based small sample classification methods. As for DFSL + SVM, RN-FSC, and UM^{2}L, four precollected HSIs including Houston 2013, Botswana, Kennedy Space Center, and Chikusei, are used as the labeled source samples. DCFSL only uses the Chikusei dataset for meta-training.

The quantitative evaluation results of different methods on the four HSIs are shown in Table IV. For fairness, we run ten experimental results for each classification method. In Table V, overall classification accuracy (OA), average classification accuracy (AA), and Kappa coefficient (kappa) are used as quantitative evaluation indicators. According to the experimental results in Table V, we can obtain the following observations.

1) The classification performance of 3D-CNN is far inferior to that of the semi-supervised method EMP-GCN. This is because 3D-CNN only uses supervised samples for end-to-end training, without utilizing unsupervised samples.
2) Compared with 3D-CNN and EMP + GCN, the classification accuracy of the four meta-learning-based methods is further improved. Meta-learning-based methods can use the additional knowledge from the source samples to guide the classification process on the target HSIs, resulting in a better classification performance.
3) Compared with the comparison methods, the classification accuracy of the DF + SVM proposed in this article is lower than that of RN-FSC and DCFSL on the PC dataset, but it still provides competitive results. On the UP, SA, and IP datasets, the classification accuracy of

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TABLE V

CLASSIFICATION RESULTS OF DIFFERENT METHODS IN THE CROSS-DOMAIN SCENARIO. SD DENOTES THE STANDARD DEVIATION OF TEN EXPERIMENTAL RESULTS

HSI	Criteria	3D-CNN *Mean±SD*	EMP+GCN *Mean±SD*	DFSL+SVM *Mean±SD*	RN-FSC *Mean±SD*	DCFSL *Mean±SD*	UM^{2}L *Mean±SD*	ADAFSL *Mean±SD*	CTF-SSCL *Mean±SD*	DF+SVM *Mean±SD*
UP	OA	66.88±2.82	77.88±1.15	75.61±4.56	80.16±2.40	80.65±4.68	80.92±3.55	85.12±1.99	82.90±3.26	84.33±2.06
	AA	61.89±3.18	76.48±0.73	75.91±2.92	78.36±3.38	81.37±1.68	77.77±3.19	85.50±1.78	85.67±1.92	89.34±1.45
	kappa	57.26±3.27	70.37±1.41	69.09±5.06	74.23±2.80	74.99±5.27	75.56±4.06	80.59±2.41	77.99±3.82	79.82±0.25
PC	OA	80.91±5.09	95.02±0.94	94.90±1.97	95.93±1.16	97.45±0.55	94.27±1.43	94.55±0.43	93.98±0.87	95.40±0.76
	AA	64.06±5.22	88.66±1.24	86.41±2.33	87.94±2.41	94.25±0.94	84.85±2.93	90.52±0.76	90.25±1.78	90.56±1.35
	kappa	73.98±6.57	93.01±1.29	92.89±2.62	94.27±1.60	96.40±0.77	91.94±1.99	92.54±0.61	93.33±1.22	93.52±0.11
SA	OA	80.12±3.08	83.05±1.47	85.31±3.47	84.54±1.77	89.13±1.69	90.34±2.54	91.43±0.47	89.85±1.60	94.46±0.65
	AA	82.62±2.19	83.61±1.63	91.23±1.11	89.47±1.36	94.02±0.72	93.04±1.30	95.28±0.58	93.97±1.18	95.35±0.94
	kappa	78.02±3.42	85.94±1.26	83.71±3.78	82.88±1.94	87.94±1.84	89.29±2.80	90.49±0.52	88.74±1.70	93.85±0.73
IP	OA	54.46±2.84	61.78±2.77	63.11±3.13	62.47±1.78	64.58±1.89	72.09±2.20	71.91±3.15	71.51±3.69	81.06±2.11
	AA	51.99±3.16	56.64±2.06	62.23±4.09	62.70±1.03	76.36±1.76	64.84±3.17	81.54±1.49	82.23±1.75	87.79±1.14
	kappa	49.54±2.80	57.78±2.92	58.53±3.20	58.25±1.87	60.10±2.04	68.83±2.37	68.23±3.39	67.90±3.98	78.59±0.23

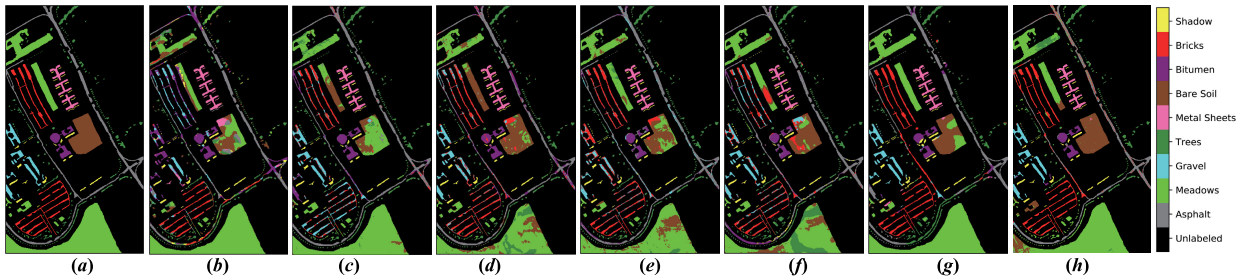


Fig. 8. Classification maps resulting from different methods on the UP dataset. (a) Ground truth. (b) 3D-CNN. (c) EMP + GCN. (d) DFSL + SVM. (e) RN-FSC. (f) DCFSL. (g) UM^{2}L. (h) DF + SVM.

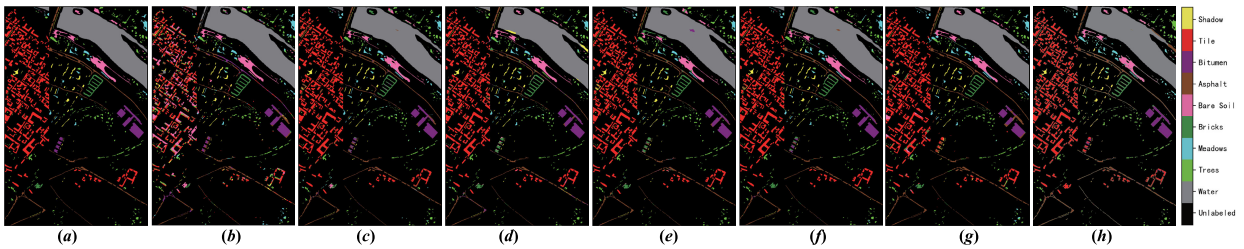


Fig. 9. Classification maps resulting from different methods on the PC dataset. (a) Ground truth. (b) 3D-CNN. (c) EMP + GCN. (d) DFSL + SVM. (e) RN-FSC. (f) DCFSL. (g) UM^{2}L. (h) DF + SVM.

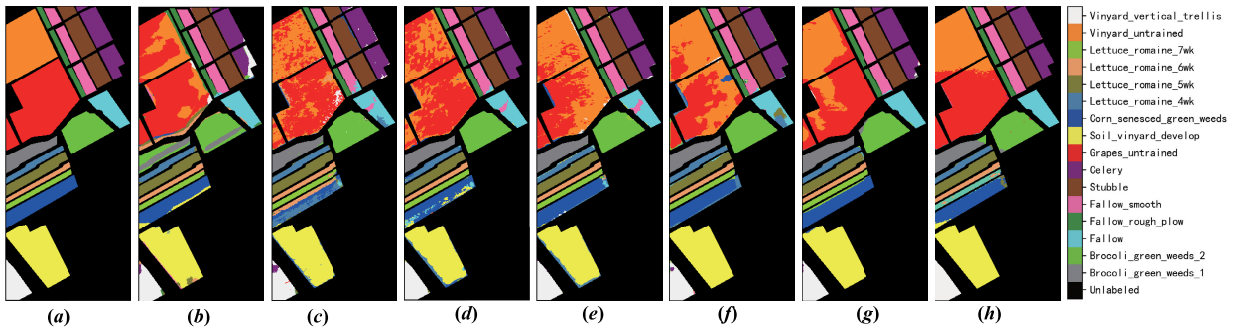


Fig. 10. Classification maps resulting from different methods on the SA dataset. (a) Ground truth. (b) 3D-CNN. (c) EMP + GCN. (d) DFSL + SVM. (e) RN-FSC. (f) DCFSL. (g) UM^{2}L. (h) DF + SVM.

DF + SVM is far higher than that of the comparison methods. For example, on the IP dataset, the OA of DF + SVM is improved by 8.97.

Compared with the comparison methods, DF + SVM does not require any training on the target data or other source data and has the feature of plug and play. The improvement in classification accuracy of DF + SVM is mainly due to the collaboration of multiple sources of information. DF + SVM uses not only spectral information, but also depth information. Unlike traditional multisource remote sensing image collaborative classification models, the depth information used by DF + SVM comes from model generation.

To directly observe the classification results, the classification maps of different methods are shown in Figs. 8–11.

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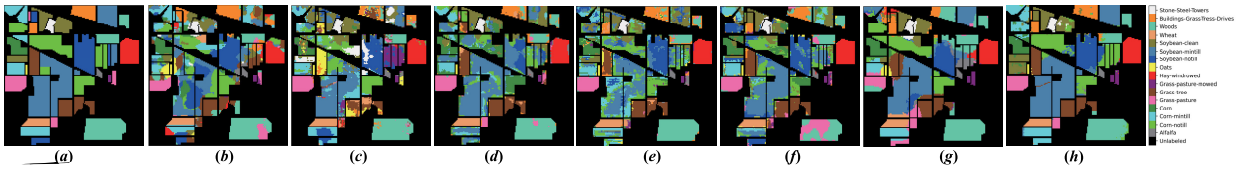


Fig. 11. Classification maps resulting from different methods on the IP dataset. (a) Ground truth. (b) 3D-CNN. (c) EMP + GCN. (d) DFSL + SVM. (e) RN-FSC. (f) DCFSL. (g) UM^{2}L. (h) DF + SVM.

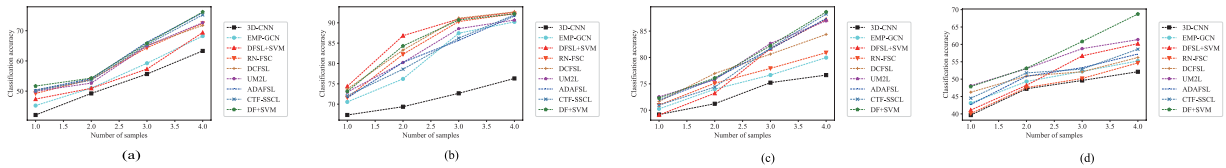


Fig. 12. Classification accuracy with different numbers of labeled samples. (a) UP. (b) PC. (c) SA. (d) IP.

TABLE VI
CLASSIFICATION ACCURACY OF DIFFERENT METHODS ON THE HOUSTON 2013 DATASET

Class No.	SVM	TBCNN	MML	CNN-DF	HRWN	DHViT	DF+SVM
C1	97.76	99.89	99.54	98.67	98.39	99.23	99.92
C2	98.09	96.90	99.75	99.86	99.40	99.47	99.68
C3	99.71	100	100	98.11	99.80	99.98	100
C4	98.96	98.46	99.21	99.62	99.91	99.36	99.92
C5	96.78	98.69	99.96	99.75	99.36	99.87	99.68
C6	98.15	97.92	99.50	98.75	99.44	99.77	100
C7	82.41	92.24	98.83	95.34	99.81	99.01	99.68
C8	86.66	91.22	99.32	95.95	99.05	99.91	99.76
C9	86.58	97.58	96.72	95.14	95.47	98.62	97.76
C10	93.81	85.80	99.50	94.56	98.55	99.97	100
C11	90.45	92.36	99.84	96.62	99.32	99.79	99.92
C12	87.27	81.34	98.22	94.62	98.36	99.70	99.92
C13	68.44	84.68	99.54	93.86	99.48	99.34	95.95
C14	98.83	90.81	99.86	96.28	99.31	99.94	100
C15	99.09	99.58	99.93	100	99.49	100	99.85
OA	92.15	93.50	99.17	97.09	99.01	99.55	99.55
AA	92.20	93.83	99.17	97.14	99.01	99.60	99.47
kappa	91.52	92.94	99.10	96.84	99.04	99.51	99.52

TABLE VII
CLASSIFICATION ACCURACY OF DIFFERENT METHODS ON THE TRENTO DATASET

Class No.	SVM	TBCNN	MML	CNN-DF	HRWN	DHViT	DF+SVM
C1	70.87	96.01	99.10	97.33	99.17	99.88	99.98
C2	95.53	88.64	96.87	96.80	99.08	98.23	98.14
C3	89.25	98.02	89.05	99.52	99.32	97.53	99.58
C4	99.88	99.69	99.97	99.96	99.08	100	100
C5	96.19	99.43	99.68	99.45	99.20	99.98	100
C6	95.70	93.52	98.29	98.35	97.01	98.51	98.30
OA	92.69	97.36	99.13	98.93	99.09	99.58	99.63
AA	91.24	95.88	97.16	98.93	98.81	99.02	99.33
kappa	90.32	96.47	98.83	98.57	99.01	99.43	99.51

We can find that methods with higher classification accuracy can produce more accurate classification maps, which further validates the effectiveness of the proposed method.

In order to further verify the effectiveness of the proposed method for small sample classification, we randomly selected 1–4 labeled samples per class on the four datasets for classification experiments. The AA of the ten classification results is shown in Fig. 12. According to the experimental results in Fig. 12, we found that our method can still obtain higher classification accuracy than the comparison method under the condition of fewer labeled samples, which further verified the effectiveness of the proposed method.

D. Comparison With Collaborative Classification Methods

The accuracy improvement of the proposed method is mainly due to the collaborative use of HSIs and depth information. To further demonstrate the effectiveness of the proposed method, five hyperspectral and LiDAR collaborative classification models, TBCNN [50], MML [51], CNN-DF [52], HRWN [50], and DHViT [53] are selected for comparative experiments on the Houston 2013 and Trento dataset. TBCNN is a two-branch CNN model that could use multimodality remote sensing data for classification. MML is a general multimodal deep learning framework for remote sensing image classification applications. CNN-DF and HRWN

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are two coupled CNNs with feature-level and decision-level fusion strategies. DHViT is the latest deep hierarchical ViT architecture for HSI and LiDAR data joint classification. To facilitate comparative analysis, we also give the experimental results of spectral information classification using SVM. For fairness, the training samples and test samples on the two datasets of Houston 2013 and Trento are consistent with those in [53].

Tables VI and VII report the OA, AA, kappa, and individual class accuracy of different methods on the two datasets. According to the experimental results in Tables VI and VII, compared with SVM using only spectral features, TBCNN, MML, CNN-DF, HRWN, and DHViT, which comprehensively use real HSI and LiDAR data for collaborative classification, can significantly improve the classification accuracy. The proposed method can provide competitive experimental results. It should be emphasized that TBCNN, MML, CNN-DF, HRWN, and DHViT all use real HSI and LiDAR data for collaborative classification. In practical application, it is difficult to obtain multisource remote sensing images of the same region at the same time, and multisource images need to be registered. In contrast, the proposed method can obtain higher classification accuracy only using the HSI and depth information generated by model, which further demonstrates the effectiveness of the proposed method.

E. Discussion

The improvement of the proposed method for small sample classification of HSIs is mainly due to the introduction of depth information. Unlike traditional multisource remote sensing image fusion classification methods, the proposed method does not require obtaining actual depth information. The proposed method obtains depth information through a pretrained foundational model. Furthermore, we propose extracting depth information of different bands along the spectral dimension to obtain more discriminative features. Finally, we combine the proposed method with SVM to complete HSI classification. A large number of classification experiments have verified the effectiveness of the proposed method. The proposed method can also be easily combined with deep learning methods. However, the proposed method requires training a depth estimation model in a semi-supervised manner, which undoubtedly increases the time cost of the proposed method.

IV. Conclusion

Aiming at the difficulty of data acquisition in the multisource remote sensing image collaborative classification, we propose to use a pretrained monocular image depth estimation model to extract the depth information of HSIs. Further, we consider the depth map extracted from different band images as features and combine it with the original spectral features to complete the final classification. The proposed method has the advantages of plug and play, and can obtain the classification accuracy equivalent to that of multisource data without additional data. In addition, the proposed method can obtain competitive classification accuracy under the condition of small samples. A large number of classification experiments on several international benchmark data verify the effectiveness of the proposed method.

References

[1] X. Cui, Z. Zhang, B. Zhao, X.-Y. Cheng, and Z.-M. Qi, "Structural characterization and optical properties of perovskite crystals based on micro-hyperspectral imaging sensor," *IEEE Sensors J.*, vol. 24, no. 1, pp. 806–813, Jan. 2024.

[2] C. Fu, B. Du, and L. Zhang, "Do we need learnable classifiers? A hyperspectral image classification algorithm based on attention-enhanced ResBlock-in-ResBlock and ETF classifier," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505013.

[3] B. Liu, X. Yu, P. Zhang, A. Yu, Q. Fu, and X. Wei, "Supervised deep feature extraction for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 1909–1921, Apr. 2018.

[4] L. Zhang, L. Zhang, D. Tao, and X. Huang, "Tensor discriminative locality alignment for hyperspectral image spectral–spatial feature extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 1, pp. 242–256, Jan. 2013.

[5] L. Zhang, Q. Zhang, B. Du, X. Huang, Y. Y. Tang, and D. Tao, "Simultaneous spectral–spatial feature selection and extraction for hyperspectral images," *IEEE Trans. Cybern.*, vol. 48, no. 1, pp. 16–28, Jan. 2018.

[6] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019.

[7] L. He, C. Liu, J. Li, Y. Li, S. Li, and Z. Yu, "Hyperspectral image spectral–spatial-range Gabor filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4818–4836, Jul. 2020.

[8] Y. Li, H. Tang, W. Xie, and W. Luo, "Multidimensional local binary pattern for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5505113.

[9] N. Falco, J. A. Benediktsson, and L. Bruzzone, "Spectral and spatial classification of hyperspectral images based on ICA and reduced morphological attribute profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 11, pp. 6223–6240, Nov. 2015.

[10] B. Liu, Y. Sun, A. Yu, Z. Xue, and X. Zuo, "Hyperspectral meets optical flow: Spectral flow extraction for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 32, pp. 5181–5196, 2023.

[11] G. Tang, X. Wang, H. Zhao, X. Hu, G. Jin, and Y. Zhong, "Attention in attention for hyperspectral with high spatial resolution (H) image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5501113.

[12] J. Yue, W. Zhao, S. Mao, and H. Liu, "Spectral–spatial classification of hyperspectral images using deep convolutional neural networks," *Remote Sens. Lett.*, vol. 6, no. 6, pp. 468–477, Jun. 2015.

[13] C. Yu, R. Han, M. Song, C. Liu, and C.-I. Chang, "A simplified 2D–3D CNN architecture for hyperspectral image classification based on spatial–spectral fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2485–2501, 2020.

[14] B. Liu, X. Yu, A. Yu, P. Zhang, and G. Wan, "Spectral–spatial classification of hyperspectral imagery based on recurrent neural networks," *Remote Sens. Lett.*, vol. 9, no. 12, pp. 1118–1127, Dec. 2018.

[15] L. Song, Z. Feng, S. Yang, X. Zhang, and L. Jiao, "Interactive spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 9, pp. 8589–8601, Sep. 2024.

[16] L. Sun, H. Zhang, Y. Zheng, Z. Wu, Z. Ye, and H. Zhao, "MASSFormer: Memory-augmented spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516415.

[17] K. He, Y. Cai, S. Peng, and M. Tan, "A diffusion model-assisted multiscale spectral attention network for hyperspectral image super-resolution," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 8612–8625, 2024.

[18] H. Lu et al., "Semi-supervised dynamic ensemble learning with balancing diversity and consistency for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5514416.

[19] H. Song and W. Yang, "SRSN: A semi-supervised robust self-ensemble network for hyperspectral images classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[20] R. Chen et al., "Prototype-based pseudo-label refinement for semi-supervised hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

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[21] J. Zhang, L. Liu, R. Zhao, and Z. Shi, "A Bayesian meta-learning-based method for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500613

[22] K. Gao, B. Liu, X. Yu, and A. Yu, "Unsupervised meta learning with multiview constraints for hyperspectral image small sample set classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3449–3462, 2022.

[23] B. Liu, A. Yu, X. Yu, R. Wang, K. Gao, and W. Guo, "Deep multiview learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7758–7772, Sep. 2021.

[24] Z. Xue, B. Liu, A. Yu, X. Yu, P. Zhang, and X. Tan, "Self-supervised feature representation and few-shot land cover classification of multimodal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541618.

[25] Z. Xue, X. Yu, B. Liu, X. Tan, and X. Wei, "HResNetAM: Hierarchical residual network with attention mechanism for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3566–3580, 2021.

[26] P. Ghamisi, J. A. Benediktsson, and S. Phinn, "Land-cover classification using both hyperspectral and LiDAR data," *Int. J. Image Data Fusion*, vol. 6, no. 3, pp. 189–215, Jul. 2015.

[27] W. Li, J. Wang, Y. Gao, M. Zhang, R. Tao, and B. Zhang, "Graph-feature-enhanced selective assignment network for hyperspectral and multispectral data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526914.

[28] J. Wang, W. Li, Y. Gao, M. Zhang, R. Tao, and Q. Du, "Hyperspectral and SAR image classification via multiscale interactive fusion network," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 12, pp. 10823–10837, Dec. 2023.

[29] L. Sun, X. Wang, Y. Zheng, Z. Wu, and L. Fu, "Multiscale 3-D–2-D mixed CNN and lightweight attention-free transformer for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 2100116.

[30] Y. Chen, C. Li, P. Ghamisi, X. Jia, and Y. Gu, "Deep fusion of remote sensing data for accurate classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1253–1257, Aug. 2017.

[31] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

[32] H. Li, W. Wang, L. Pan, W. Li, Q. Du, and R. Tao, "Robust capsule network based on maximum correntropy criterion for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 738–751, 2020.

[33] H. Li, P. Ghamisi, U. Soergel, and X. Zhu, "Hyperspectral and LiDAR fusion using deep three-stream convolutional neural networks," *Remote Sens.*, vol. 10, no. 10, p. 1649, Oct. 2018.

[34] D. Wang, Z. Liu, S. Shao, X. Wu, W. Chen, and Z. Li, "Monocular depth estimation: A survey," in *Proc. 49th Annu. Conf. IEEE Ind. Electron. Soc.*, Oct. 2023, pp. 1–7.

[35] D. Eigen, C. Puhrsch, and R. Fergus, "Depth map prediction from a single image using a multi-scale deep network," in *Proc. Neural Inf. Process. Syst.*, Jun. 2014, pp. 2366–2374.

[36] A. Johnston and G. Carneiro, "Self-supervised monocular trained depth estimation using self-attention and discrete disparity volume," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4755–4764.

[37] Y. Kuznietsov, J. Stückler, and B. Leibe, "Semi-supervised deep learning for monocular depth map prediction," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2215–2223.

[38] B. Liu, X. Chen, D. Zhou, P. Wang, and R. Wang, "Text guided zero-shot scene classification of high spatial resolution remote sensing images," *J. Appl. Remote Sens.*, vol. 18, no. 1, Mar. 2024, Art. no. 014525.

[39] A. Kirillov et al., "Segment anything," 2023, *arXiv:2304.02643*.

[40] L. Yang, B. Kang, Z. Huang, X. Xu, J. Feng, and H. Zhao, "Depth anything: Unleashing the power of large-scale unlabeled data," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 10371–10381.

[41] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. ICLR*, Jan. 2020, pp. 1–21.

[42] S. Yun, D. Han, S. J. Oh, S. Chun, J. Choe, and Y. Yoo, "CutMix: Regularization strategy to train strong classifiers with localizable features," 2019, *arXiv:1905.04899*.

[43] M. Caron et al., "Emerging properties in self-supervised vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9630–9640.

[44] B. Liu, X. Yu, P. Zhang, X. Tan, R. Wang, and Z. Lu, "Spectral–spatial classification of hyperspectral image using three-dimensional convolution network," *J. Appl. Remote Sens.*, vol. 12, no. 1, Jan. 2018, Art. no. 016005.

[45] B. Liu, K. Gao, A. Yu, W. Guo, R. Wang, and X. Zuo, "Semisupervised graph convolutional network for hyperspectral image classification," *J. Appl. Remote Sens.*, vol. 14, no. 2, Jun. 2020, Art. no. 026516.

[46] K. Gao, B. Liu, X. Yu, J. Qin, P. Zhang, and X. Tan, "Deep relation network for hyperspectral image few-shot classification," *Remote Sens.*, vol. 12, no. 6, p. 923, Mar. 2020.

[47] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501618.

[48] Z. Ye, J. Wang, H. Liu, Y. Zhang, and W. Li, "Adaptive domain-adversarial few-shot learning for cross-domain hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5532017.

[49] B. Xi, Y. Zhang, J. Li, Y. Li, Z. Li, and J. Chanussot, "CTF-SSCL: CNN-transformer for few-shot hyperspectral image classification assisted by semisupervised contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5532617.

[50] X. Zhao et al., "Joint classification of hyperspectral and LiDAR data using hierarchical random walk and deep CNN architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7355–7370, Oct. 2020.

[51] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.

[52] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[53] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

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